Sai

UCSH-201

AVL TREE

→ Adelson-Velskii & Landis

→ BST + Balance criteria

→ $|H_L - H_R| \leq 1$

→ 4 Cases of imbalance

- → L S L H (Right Rotation)
- → L S R H (Left Rot. on Left Child + Right Rotation)
- → R S L H (Right Rot. on Right Child + Left Rotation)
- → R S R H (Left Rotation)

→ Implementation !

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```
struct Node
{
    int val;
    struct Node *l, *r;
    int h;
};

int height(struct Node *root)
{
    if (root == NULL)
        return (-1);
    return (root->h);
}

int max(int x, int y)
{
    if (x > y)
        return(x);
    return(y);
```

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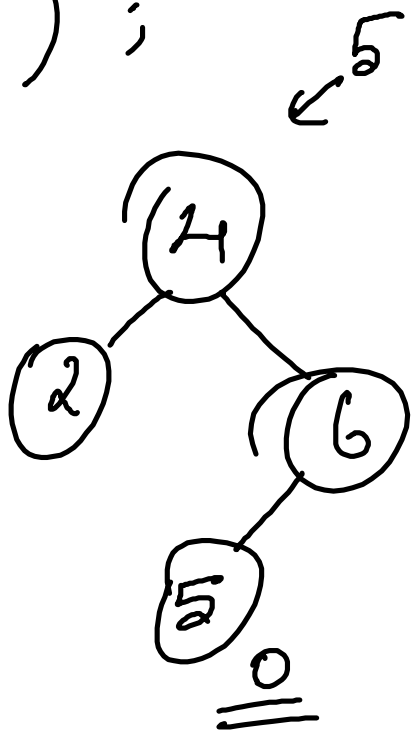
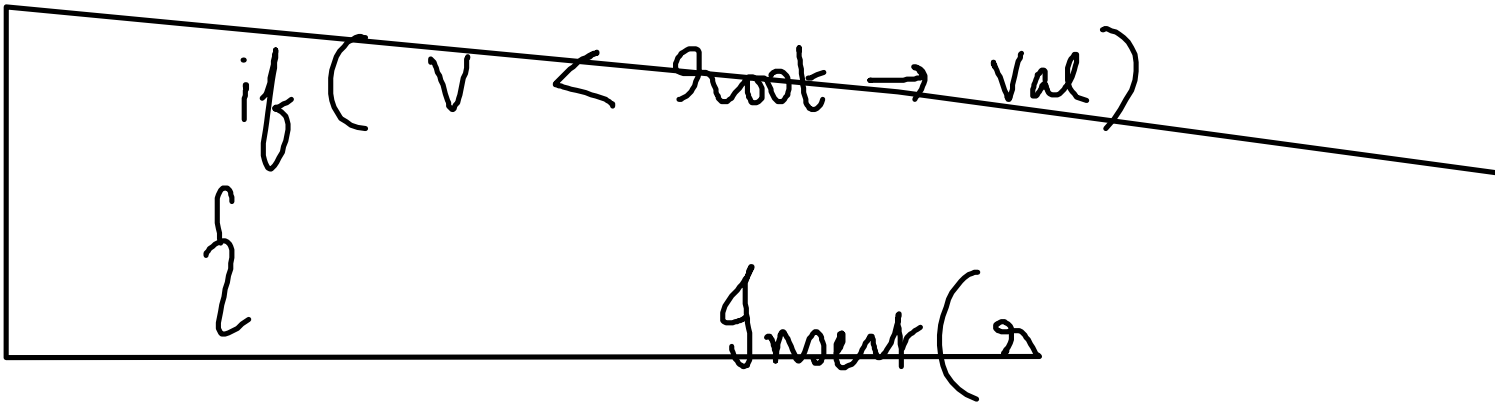
```
struct Node * createNode(int v)
{
    struct Node *n = NULL;
    n = (struct Node *) malloc(
            sizeof(struct Node));
    n -> val = v;
    n -> l = NULL;
    n -> r = NULL;
    n -> h = 0;
    return (n);
}
```



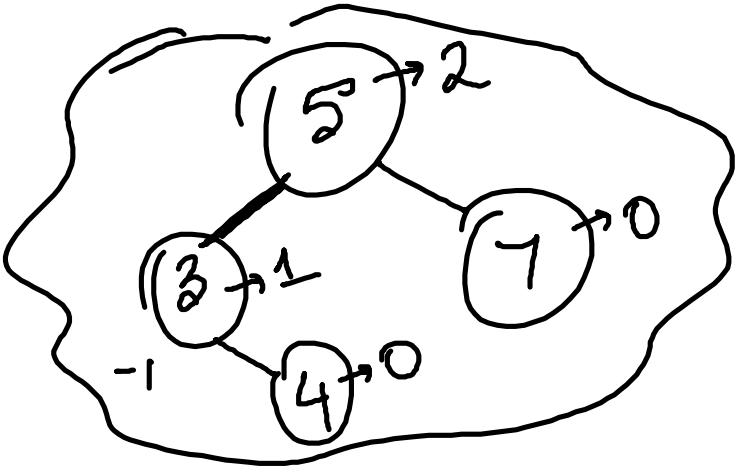

```
struct Node * Insert(struct Node *root,
                     int v)
{
    if (root == NULL)
        return (createNode(v));
```


if (v < root → val)
{
Insert(

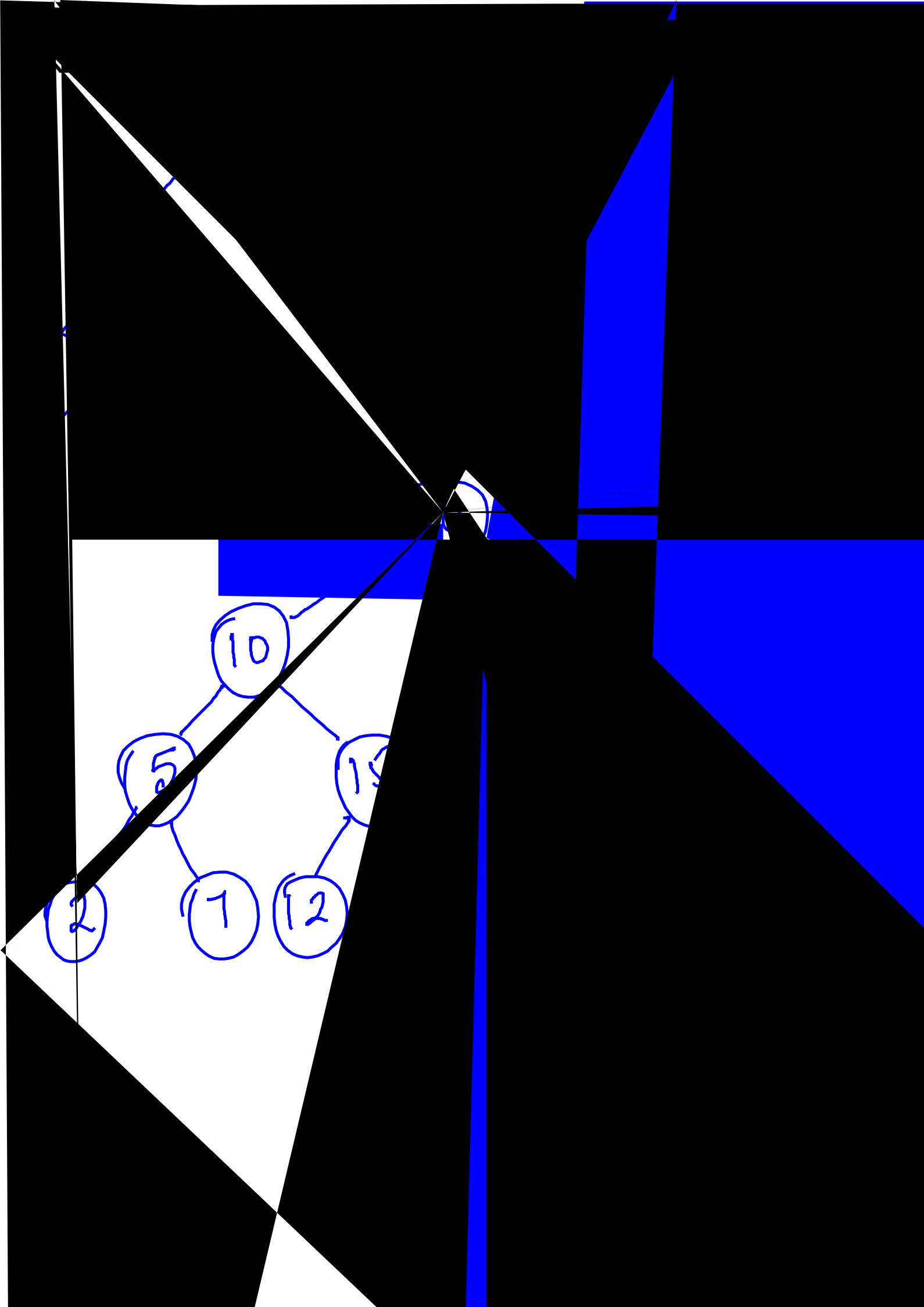
```
else if ( v > root -> val)
{
    root -> r = Insert(root -> r, v);
    if ((height (root -> r) - height(root -> l))
            == 2)
    {
        if ( v > root -> r -> val)
            // R S R H
            return ( LeftRotate (root));

        else if ( v < root -> r -> val)
            // R S L H
            return(RLRotate(root));
    }
}
```

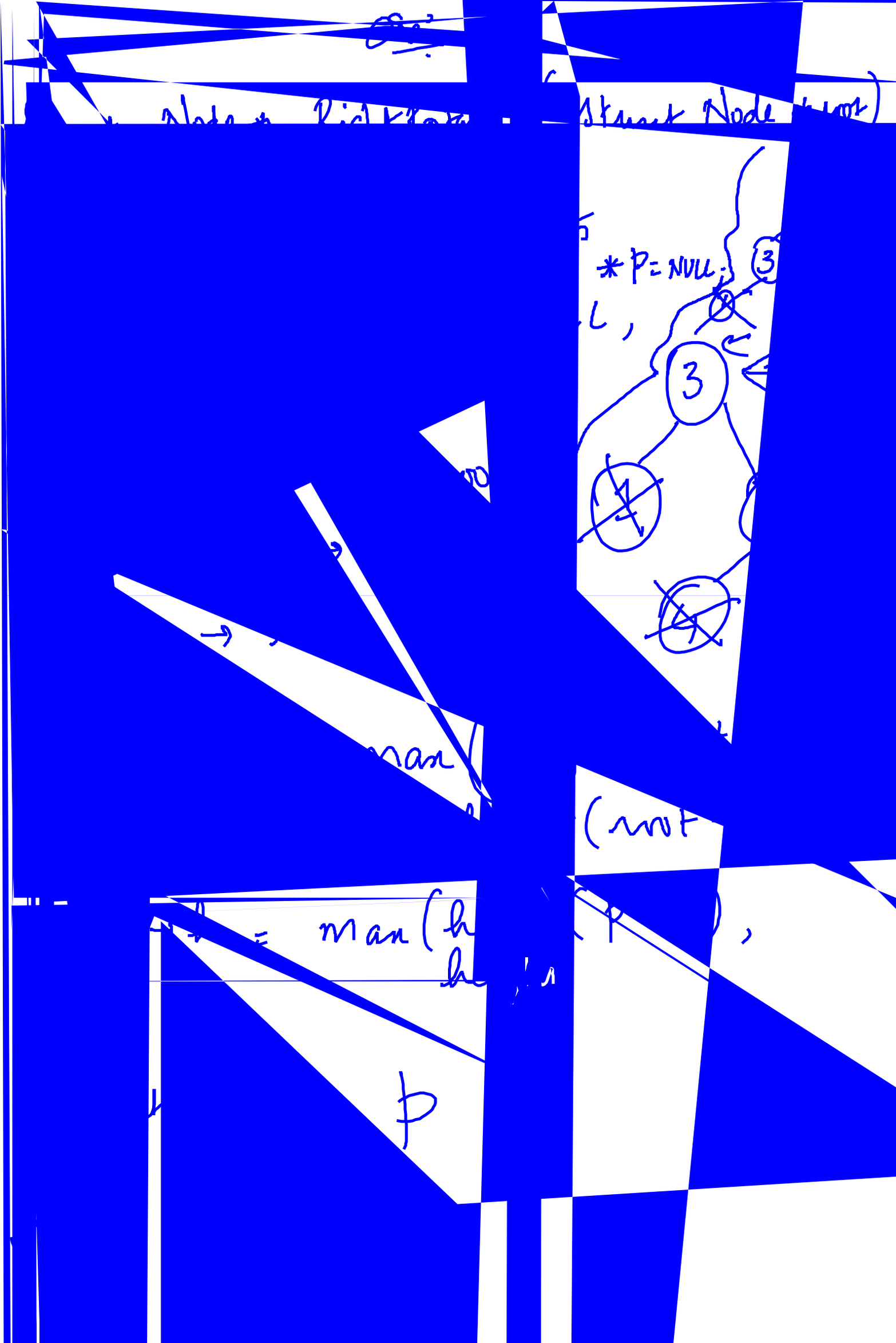
```
root -> h = max(height(root -> l),
                height(root -> r)) + 1;

return (root);
}
```

max(0,0)




```
int main()
{
    struct Node *root = NULL;
    root = Insert(root, 5);
    "      "  (  "  , 7);
    "      "  "  (  "  , 3);
    "      "  "  (  "  , 4);
}
```


10
5
2
7
12


*p=NULL;
3
3

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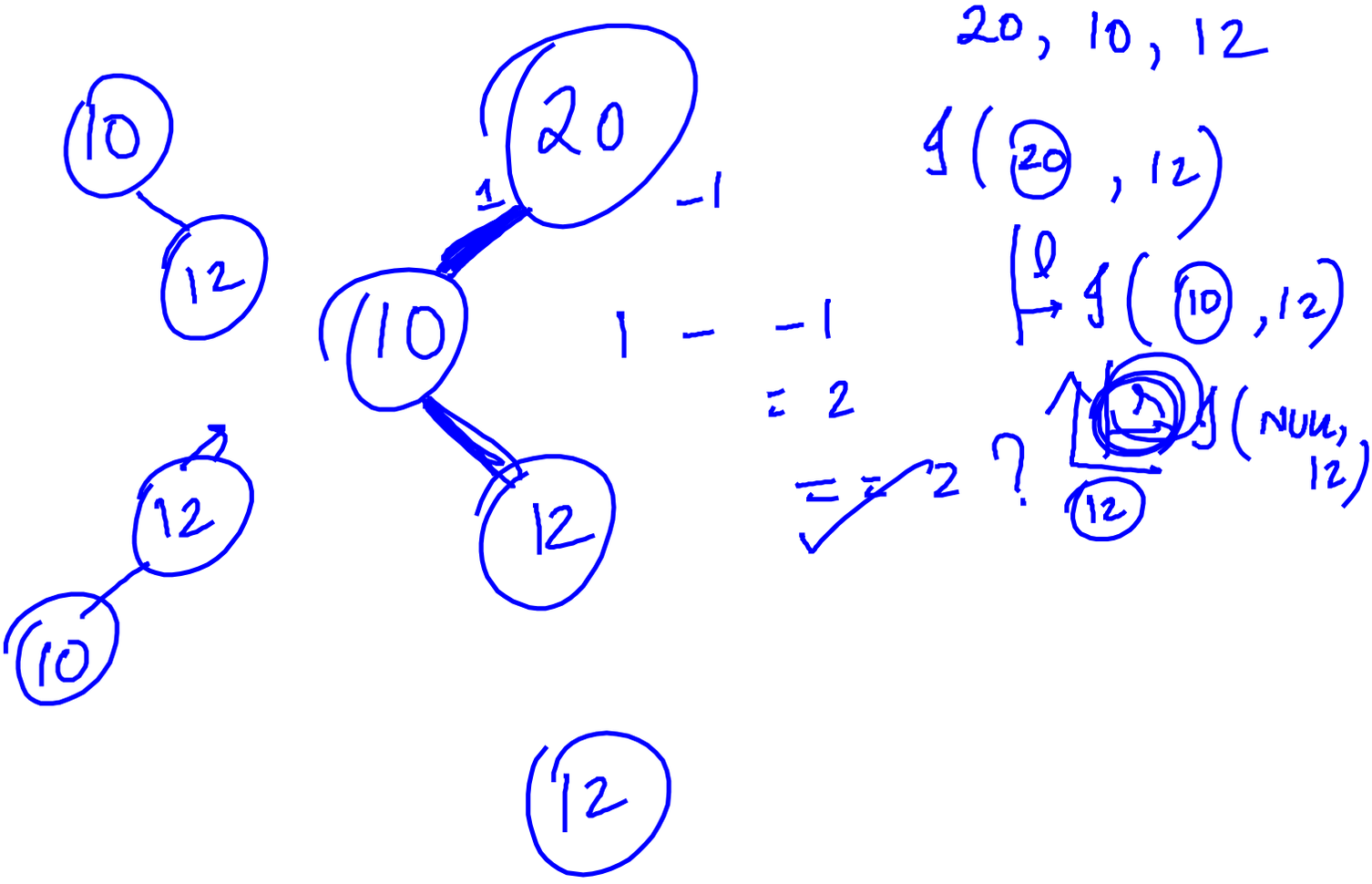
Struct Node * LR Rotate (Struct Node *root)

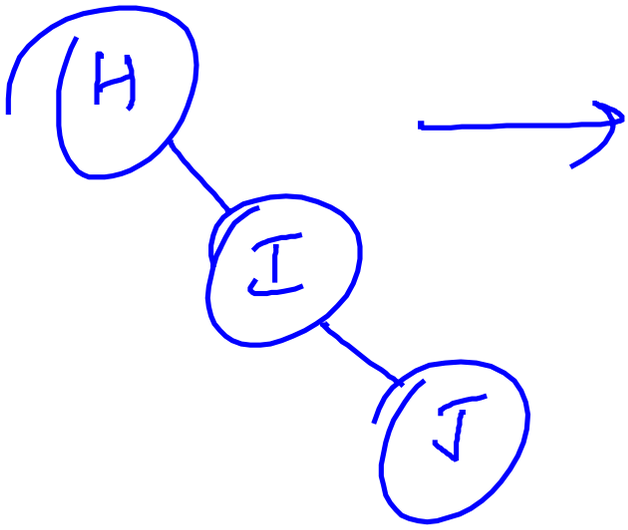
{ st Perform Left Rotate on the child, followed by Right Rotate the node of imbalance.

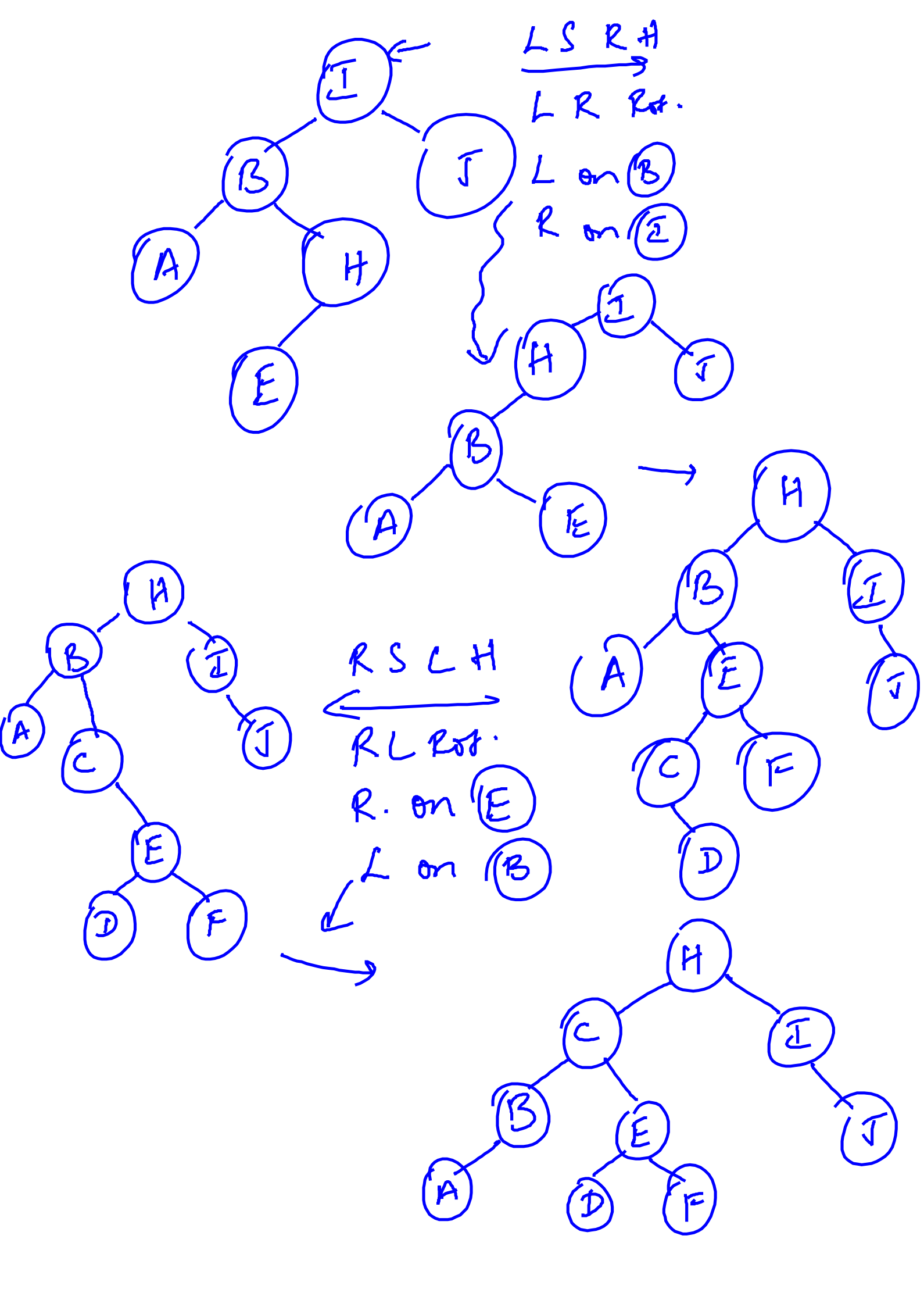
= Left Rotate (root → l);

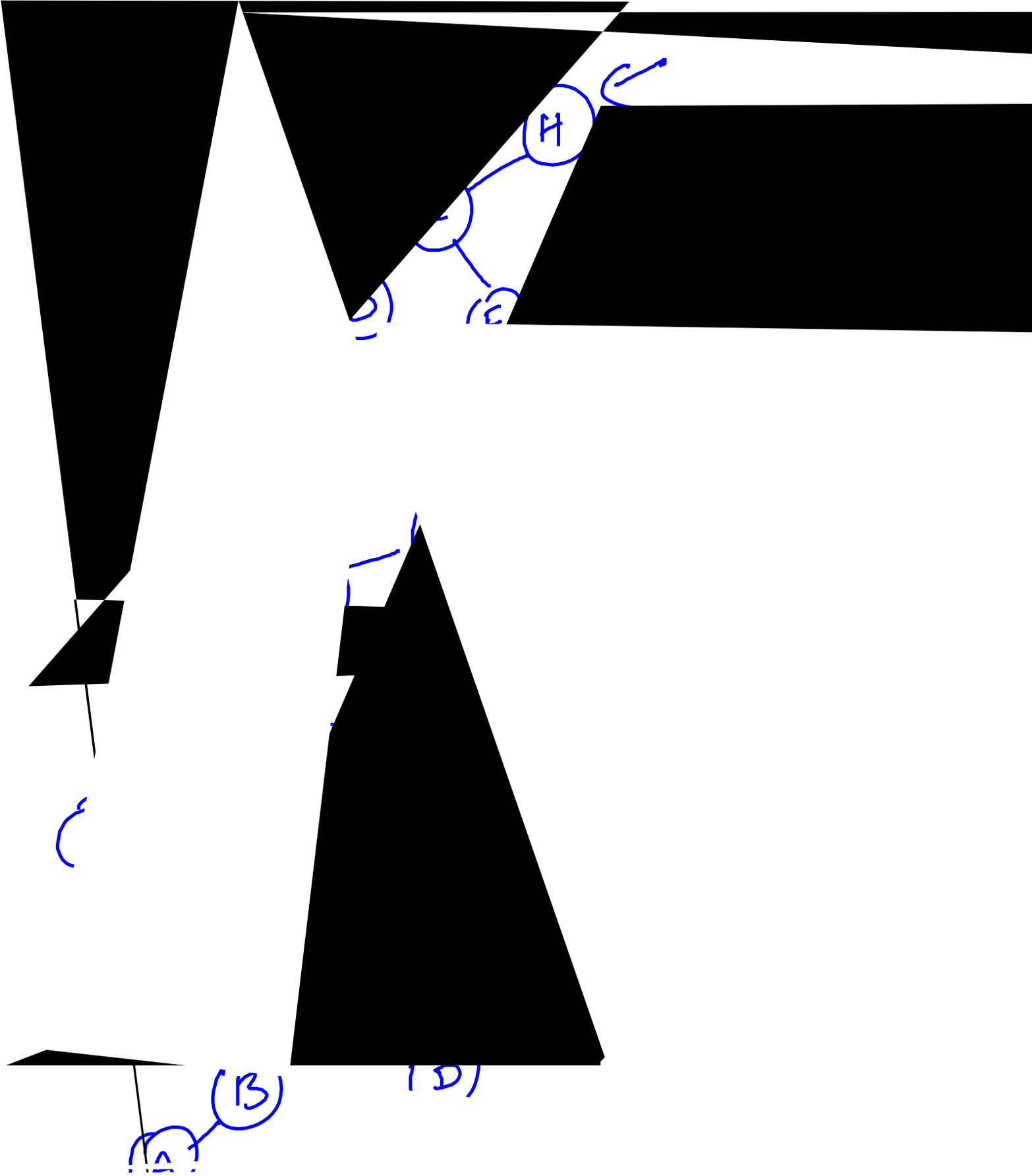
Rotate (root);



H, I, J, B, A, E, C, F, D, G, K, L




LSRH
LR Rot.
L on B
R on I
RSLH
RL Rot.
R. on E
L on B


H
B